Learn 2D, Solve 3D: UA-Fit, an Uncertainty-Weighted Analytical Solver for Multi-view Hand Mesh Recovery Supplementary Material

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A Derivations and Algorithms

This supplementary material is organized in two parts. Sec. A expands the closed forms, Jacobians, and propagation steps quoted compactly in the main paper. None introduces trainable parameters. Sec. B reports the extended experiments: the view-count sweep, the statistical reliability of the main paper's Table 1, solver refinements, the inference-time anatomical energy, the backward-mode study, and batch-scaling efficiency. Algorithm 1 states the complete inference procedure, whose individual steps the subsections below derive.

A.1 Anisotropic covariance propagation ($\boldsymbol{\Omega}^2 \to \boldsymbol{\Omega}^3$)

Triangulation in the main paper is a weighted nonlinear least-squares problem whose measurement model is $\mathbf{u}_{vj} = \pi_v(\mathbf{x}_j) + \boldsymbol{\varepsilon}_{vj}$ with $\mathrm{Cov}[\boldsymbol{\varepsilon}_{vj}] = \boldsymbol{\Omega}_{vj}^{-1}$. Linearizing at the optimum $\boldsymbol{\mu}_j$ with $J_{vj} = \partial\pi_v/\partial\mathbf{x}$, the maximum- likelihood (Gauss–Newton) estimator has first-order covariance

$$\boldsymbol{\Sigma}_j \;=\; \Big(\textstyle\sum_v J_{vj}^\top \boldsymbol{\Omega}_{vj} J_{vj}\Big)^{-1}, \tag{1}$$

i.e. the inverse GN Hessian returned by the last triangulation step. It is anisotropic and depth-aware: at low view counts it elongates along the dominant baseline. Its inverse $\boldsymbol{\Omega}_j^3 = \boldsymbol{\Sigma}_j^{-1}$ is the precision used to weight the 3D anchor in the main paper's fitting energy.

A.2 Robust IRLS weights

The redescending weight $w(\rho) = c^2/(c^2+\rho)$ applied to the Mahalanobis residual $\rho_{vj} = \mathbf{r}^\top \boldsymbol{\Omega}\mathbf{r}$ is the IRLS weight of the Cauchy (Lorentzian) M-estimator $\rho_{\mathrm{C}}(e) =$

Algorithm 1 UA-Fit inference. The 3D estimation (triangulation, Kabsch, and LM fit) is parameter-free. The only learned step is the feed-forward shape head $\hat{\boldsymbol{\beta}}$, which is counted in the reconstruction stage.

Require: crops $\{I_v\}$, intrinsics $\{K_v\}$, extrinsics $\{T_v\}$, MANO

$f_v, L_v \leftarrow$ HRNet+heads$(I_v)\ \forall v$ $\triangleright$ DOVF field, Cholesky field (learned)

$\mathbf{u}_{vj}, \boldsymbol{\Omega}_{vj} \leftarrow$ vote$(f_v[j], L_v[j])$ $\triangleright$ 2D consensus + aggregated precision

$\boldsymbol{\mu}_j \leftarrow$ DLT$(\{\mathbf{u}_{vj}\})$ $\triangleright$ SVD init

for $k = 1..2$ **do** $\triangleright$ $\boldsymbol{\Omega}$-weighted GN triangulation

  $A_j \leftarrow \sum_v J_{vj}^\top \boldsymbol{\Omega}_{vj} J_{vj} + \epsilon I$; $\boldsymbol{\mu}_j \leftarrow \boldsymbol{\mu}_j + A_j^{-1} \sum_v J_{vj}^\top \boldsymbol{\Omega}_{vj} \mathbf{r}_{vj}$

optional IRLS: $w_{vj} \leftarrow c^2/(c^2 + \mathbf{r}^\top \boldsymbol{\Omega} \mathbf{r})$, redo GN $\triangleright$ robust re-weighting

$\boldsymbol{\Sigma}_j \leftarrow A_j^{-1}$ $\triangleright$ closed-form 3D covariance

$(\boldsymbol{\theta}, \mathbf{t}) \leftarrow$ Kabsch(canonical joints $\rightarrow \boldsymbol{\mu}$); $\boldsymbol{\beta} \leftarrow \mathbf{0}$ $\triangleright$ global init

$\boldsymbol{\beta} \leftarrow \hat{\boldsymbol{\beta}}$ $\triangleright$ feed-forward shape head ($\sim$70k params)

$(\boldsymbol{\theta}, \mathbf{t}) \leftarrow$ LM$(E;\ \boldsymbol{\theta}, \mathbf{t} \mid \boldsymbol{\beta})$ $\triangleright$ ONE solve; Alg. 2

return $\boldsymbol{\theta}, \boldsymbol{\beta}, \mathbf{t}$ $\triangleright$ MANO-consistent hand

$\frac{c^2}{2}\log(1+e^2/c^2)$: its influence is $\psi(e) = e/(1+e^2/c^2)$, so $w = \psi(e)/e = c^2/(c^2 + e^2)$ with $e^2 = \rho$. For an inlier $\rho \sim \chi_2^2$, so c^2 is a cutoff in Mahalanobis-squared units ($c^2 = 25$ in our runs). A view whose whole hand disagrees has ρ orders of magnitude larger, so $w \to 0$ once at least two views agree. The Huber variant $w = \min(1, c^2/\rho)$ behaves identically on the inlier set, and accuracy is insensitive to the cutoff over $c^2 \in \{16, 25, 64\}$.

Its measured effect on the benchmarks is a null: removing the reweighting leaves MPVPE unchanged to within 0.1 mm on all five splits, and exactly unchanged on InterHand2.6M, the two-hand set where wrong-hand rejection should matter most. This is the expected outcome rather than a failure of the estimator: because every crop is centred on the target hand, no view observes the wrong hand, so the robust weights have nothing to reject. The reweighting is insurance for the detector-in-the-loop regime, where a crop can land on the wrong hand. That regime is outside the scope stated in the main paper's limitations, so we retain the component but do not claim accuracy from it.

A.3 Reprojection and articulated Jacobians

The LM data term differentiates the sampled field residual through projection and kinematics,

$$\frac{\partial \mathbf{r}_{vj}}{\partial \boldsymbol{\phi}} = \underbrace{\nabla_{\mathbf{u}} f_v[j]}_{2\times 2}\ \underbrace{\frac{\partial \pi_v}{\partial \mathbf{x}}}_{2\times 3}\ \underbrace{\frac{\partial \mathbf{x}_j}{\partial \boldsymbol{\phi}}}_{3\times 51}, \tag{2}$$

combining the field's spatial gradient, the pinhole projection Jacobian, and the analytic articulated (cross-product) Jacobian over the kinematic tree. With camera point $\mathbf{x}_c = R\mathbf{x} + \mathbf{t}_c$, the pinhole projection Jacobian is

$$\frac{\partial \pi_v}{\partial \mathbf{x}} = \frac{1}{z_c}\begin{bmatrix} f_x & 0 & -f_x x_c/z_c \\ 0 & f_y & -f_y y_c/z_c \end{bmatrix} R, \tag{3}$$

and $\nabla_{\mathbf{u}} f$ is the field's spatial gradient (bilinear, sampled at the current projection). The articulated term is the cross-product Jacobian over the kinematic tree with the SO(3) right-Jacobian correction: for joint i, ancestor j, and rotation component m,

$$\frac{\partial \mathbf{x}_i}{\partial \theta_{j,m}} = \big(R_j^g J_r(\boldsymbol{\theta}_j)\, \mathbf{e}_m\big) \times (\mathbf{x}_i - \mathbf{o}_j), \qquad \frac{\partial \mathbf{x}_i}{\partial \mathbf{t}} = I_3, \tag{4}$$

where $R_j^g, \mathbf{o}_j$ are joint j's global rotation and origin, and fingertips ride their distal joint's chain as rigidly skinned points. The MANO pose-corrective blend-shape term is omitted from Eq. (4): it contributes little to the Gauss–Newton direction, as the solver study in the main paper confirms.

A.4 SO(3) right Jacobian

The correction J_r in Eq. (4) is the standard closed form

$$J_r(\boldsymbol{\theta}) = I - \tfrac{1-\cos\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^2}[\boldsymbol{\theta}]_\times + \tfrac{\|\boldsymbol{\theta}\|-\sin\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^3}[\boldsymbol{\theta}]_\times^2, \tag{5}$$

with $[\cdot]_\times$ the skew operator; $J_r \to I$ as $\|\boldsymbol{\theta}\| \to 0$. It maps an axis-angle increment $\mathbf{e}_m$ to the body-frame angular velocity so that the cross-product Jacobian is exact for the exponential-map parameterization used at each joint.

A.5 Implicit backward at the LM fixed point

At convergence the fit satisfies the stationarity condition $g(\boldsymbol{\phi}^\star, \mathbf{z}) = 0$, where $\mathbf{z}$ collects the learned inputs (fields f_v, Cholesky L_v) and g is the damped normal-equation gradient. Differentiating and using $A = \partial g/\partial \boldsymbol{\phi}$ (the GN system already formed at the last iterate) gives

$$\frac{\partial \boldsymbol{\phi}^\star}{\partial \mathbf{z}} = -A^{-1}\frac{\partial g}{\partial \mathbf{z}}, \tag{6}$$

so a downstream gradient $\bar{\boldsymbol{\phi}}$ is back-propagated as the single vector–Jacobian product $-(\partial g/\partial \mathbf{z})^\top A^{-\top}\bar{\boldsymbol{\phi}}$, at $O(1)$ memory in the number of LM iterations. Stage B uses this path to backpropagate 3D supervision through the fit and train the released model. A 2D-only variant that never invokes it also works (main paper, training setup).

A.6 Block-weighted LM loop

Alg. 2 is the pose/translation fit invoked once per scene (Alg. 1, line 10): each iteration re-reads the fields at the current projection, forms the block-weighted normal equations, takes a full step, and accepts or rejects it under a per-sample trust region.

Algorithm 2 Block-weighted adaptively damped LM (one pose/trans fit)

$\boldsymbol{\phi} \leftarrow (\boldsymbol{\theta}, \mathbf{t})$; $\quad \mu \leftarrow 10^{-3}$ (per sample)
for $k = 1..15$ **do**
 $\mathbf{r}_{vj} \leftarrow f_v[j](\pi_v(\mathbf{x}_j(\boldsymbol{\phi})))$; $\quad \boldsymbol{\Omega}_{vj} \leftarrow LL^\top(\pi_v(\mathbf{x}_j)) \cdot s_{vj}$ ▷ re-read fields
 $J_{vj} \leftarrow \nabla_{\mathbf{u}} f \cdot \frac{\partial \pi_v}{\partial \mathbf{x}} \cdot \frac{\partial \mathbf{x}_j}{\partial \boldsymbol{\phi}}$ ▷ Eqs. (2)–(4), detached
 $A \leftarrow \sum_{v,j} J^\top \boldsymbol{\Omega} J + P + \mu I$; $\quad g \leftarrow \sum_{v,j} J^\top \boldsymbol{\Omega}\, \mathbf{r} + p$
 $\boldsymbol{\phi}' \leftarrow \boldsymbol{\phi} - A^{-1} g$
 if $E(\boldsymbol{\phi}') < E(\boldsymbol{\phi})$ **then** $\boldsymbol{\phi} \leftarrow \boldsymbol{\phi}'$; $\mu \leftarrow 0.3\mu$
 else $\mu \leftarrow 3\mu$ ▷ per-sample trust region

Table 1: Training-recipe ablations. Models retrained from scratch and evaluated on the full splits (MPVPE, mm). The control model is the baseline run used for this specific ablation study.

ablation	DexYCB	HO3D	ARCTIC	InterHand	OakInk
control	**7.9**	**8.3**	**7.7**	**9.7**	**9.6**
scalar-$\boldsymbol{\Omega}$ (isotropic)	7.9	8.3	8.1	9.9	9.6
NLL-off (no calibration)	7.9	8.3	8.0	9.8	9.6

B Extended Experiments

B.1 Training-Recipe Ablations

Table 1 presents the training-recipe ablations that had a measurable effect on performance. We retrain the model under various configurations and evaluate on the full test sets (MPVPE, mm). Removing the learned anisotropic precision during training (scalar-$\boldsymbol{\Omega}$) or the calibration loss (NLL-off) slightly degrades generalization (e.g., on ARCTIC).

Finally, Table 2 provides the absolute values for the Stage-A vs Stage-B progression discussed in the main paper. Training through the solver (Stage-B) improves upon the 2D-pretrained fields (Stage-A) across every benchmark.

B.2 View-count analysis and scope (main paper, efficiency and limitations)

Fig. 1 sweeps the view count against the parametric baseline and substantiates the $\geq$ 3-view scope stated in the main paper. At 2 views, POEM-v2-param's learned fusion is more accurate on the single-hand rigid sets (HO3D 13.0 vs our 13.8 mm MPVPE). This depth-along-the-baseline advantage is one our controlled study could not recover through 2-view-specialized training, larger heads, or learned priors. As views accumulate, the parameter-free triangulation catches up: on HO3D it reaches parity by 3 views and passes POEM from there, and on ARCTIC the two stay within a few tenths of a millimetre at every view count. On InterHand2.6M, UA-Fit leads at every view count. DexYCB and OakInk stay

Table 2: Stage progression. The 2D-only fields (Stage-A) already give a usable fit, but training through the solver (Stage-B) improves performance everywhere (MPVPE / MPJPE, mm).

stage	DexYCB	HO3D	ARCTIC	InterHand	OakInk
Stage-A (2D-only, $\boldsymbol{\Omega} = I$)	9.8 / 9.5	9.3 / 8.6	9.9 / 8.8	10.2 / 9.1	10.6 / 9.9
Stage-A′ (+$\boldsymbol{\Omega}$, no solver)	8.0 / 7.7	**8.1** / **7.5**	8.1 / 7.0	9.7 / 8.6	**9.6** / **8.9**
Stage-B (final)	**7.9** / **7.5**	**8.1** / **7.5**	**7.5** / **6.6**	**8.8** / **7.8**	**9.6** / **8.9**

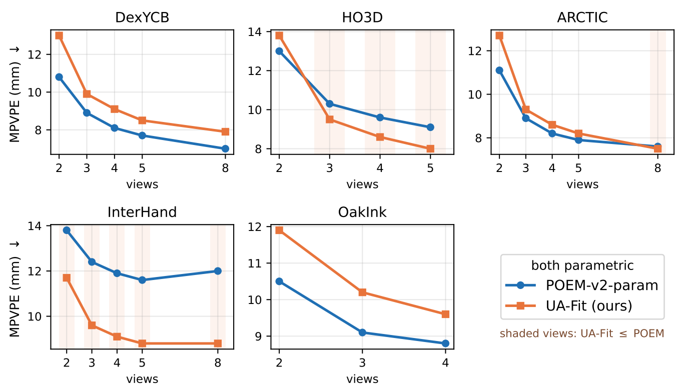

Fig. 1: MPVPE vs. view count (UA-Fit and POEM-v2-param, matched seed and cropping). Intermediate view counts use an 8000-scene sweep and the maximum-view point is the full-split value of the main paper's Table 1. Shaded = UA-Fit on-par-or-better. The parameter-free 3D solver is less accurate at 2 views (depth-bound), then matches and overtakes the learned fusion head as views accumulate, and leads at every count on InterHand, while DexYCB and OakInk stay POEM-favored, and ARCTIC stays near parity.

POEM-favored throughout, consistent with a 2D-evidence limit rather than a solver limit. We therefore scope the claim to the $\geq$ 3-view regime, trading 1–2 mm at 2 views for parametric, MANO-consistent output from a 20$\times$ smaller head.

B.3 Statistical reliability of Table 1 (main paper, setup)

Table 1 of the main paper compares the two methods under a single evaluation seed, so here we quantify how much of each reported margin could be explained by chance. First, both models are evaluated on the *complete* official splits (DexYCB 4931, HO3D 2687, ARCTIC 17359, InterHand 85235, and OakInk 21322 scenes at the fixed maximum view count), so each of its means covers every scene of its benchmark rather than a sampled subset. Second, to measure how sensitive our means are to the particular scenes in each split, we bootstrap them: each split is resampled per scene with replacement 10^4 times

Table 3: Full-split means with 95% per-scene bootstrap confidence intervals (UA-Fit, mm).

dataset	MPVPE	MPJPE	PA-MPVPE
DexYCB	7.91 [7.83, 8.00]	7.51 [7.43, 7.60]	4.86 [4.82, 4.91]
HO3D	8.05 [7.96, 8.14]	7.45 [7.37, 7.54]	5.09 [5.05, 5.13]
ARCTIC	7.54 [7.51, 7.58]	6.62 [6.58, 6.65]	5.66 [5.64, 5.69]
InterHand	8.77 [8.75, 8.78]	7.79 [7.78, 7.81]	6.23 [6.22, 6.25]
OakInk	9.58 [9.54, 9.63]	8.86 [8.82, 8.91]	5.67 [5.64, 5.69]

(seed 1), the mean error is recomputed on every resample, and the central 95% of these means is reported as a confidence interval. These intervals quantify scene-sampling variation for the fixed released checkpoints. Run-to-run training variation is characterized below with two replications, a shape-head refit and a full pipeline retrain:

Every interval half-width in Table 3 is at most ± 0.09 mm. All per-dataset margins of 0.5 mm or more in the main paper's Table 1 are therefore far larger than this scene-level noise, whereas the 0.06 mm ARCTIC MPVPE margin (7.54 vs 7.60) is not. It is also smaller than run-to-run training variation, measured at two levels. Re-fitting the shape head with a second seed (backbone, fields, and pose pathway frozen) shifts the full-split means by at most 0.1 mm on DexYCB, HO3D, ARCTIC, and OakInk and by 0.5 mm on InterHand2.6M. A full training replication with a third seed (uncertainty and fitting stages retrained under the identical recipe, shared 2D pre-training) shifts the matched-protocol means by -0.1, $+0.1$, $+0.2$, $+0.5$, and 0.0 mm on DexYCB, HO3D, ARCTIC, InterHand, and OakInk. Both leave every per-benchmark verdict of Table 1 unchanged. The main paper therefore reports ARCTIC as parity.

B.4 Solver refinements: robustness and 3D-anchor scale (main paper, solver study)

Cauchy IRLS. On the centred-hand crops of these benchmarks the robust reweighting is accuracy-neutral: removing it changes MPVPE by at most 0.1 mm on every split and leaves InterHand2.6M unchanged, because no single view observes the wrong hand. We retain it as insurance for the detector-in-the-loop regime the main paper scopes out. **3D-anchor scale** (full splits, released model, Table 1 configuration, paired per-scene bootstrap with 10^4 resamples). The dense-field data term of the fit energy is in heatmap pixels^2 through large projection Jacobians, while the anchor to the triangulated $\boldsymbol{\mu}$ is in metres^2: at the classical λ_{3D}=0.005 the anchor is 10^5–$10^7\times$ smaller than the data term and inert (all metrics identical to λ_{3D}=0). Rescaled to bind ($\lambda_{3D}\sim 10^5$), the anchor improves MPVPE on four of the five benchmarks (DexYCB +0.09, HO3D +0.19, OakInk +0.03, InterHand +0.76 mm, ARCTIC neutral at -0.01) and tightens the fit-to-$\boldsymbol{\mu}$ residual everywhere (*e.g.* HO3D $2.9 \to 2.2$ mm, InterHand $7.2 \to 6.4$ mm). Table 1 of the main paper uses this binding regime. At 2 views the picture is mixed: with two rays $\boldsymbol{\mu}$ is depth-poor and an isotropic anchor imports that depth error, so the $\boldsymbol{\Sigma}$-weighted anisotropic anchor of the fit energy is the principled low-view form.

$\boldsymbol{\Omega}$-site separation and anisotropy (main paper, weighting ablation). On the released model, seed 1, on a matched site-separation subset (distinct from the solver-study subset of the main paper, so the solver-term figure differs slightly from that of the solver table's row b): removing the learned $\boldsymbol{\Omega}$ from the solver term only costs +0.14 mm mean MPVPE (+0.2, +0.1, +0.3, +0.1, and 0.0), from the triangulation only +0.76 (+0.8, +0.4, +1.3, +0.5, and +0.8), and from both +0.98 (+1.1, +0.6, +1.5, +0.7, and +1.0; the +0.6 to +1.5 range reported in the main paper), with the two sites nearly additive. Replacing the learned triangulation by a raw Gauss–Newton triangulation that keeps the learned $\boldsymbol{\Omega}$ isolates the weighting itself: the learned refinement contributes only +0.10, so +0.66 of the triangulation-site effect is the $\boldsymbol{\Omega}$ weighting. Isotropizing the learned Cholesky field at evaluation (mean of the diagonal channels, magnitude kept, directionality discarded) changes no benchmark at 0.1 mm resolution: the inference-time value of the learned reliability is its scalar magnitude.

Uncertainty calibration. Per (view, joint), the predicted σ (from $\boldsymbol{\Sigma}{=}\boldsymbol{\Omega}^{-1}$ sampled at the consensus) correlates with the actual 2D consensus error against projected ground-truth joints at r of 0.61, 0.46, 0.69, 0.61, and 0.55 on DexYCB, HO3D, ARCTIC, InterHand, and OakInk respectively. The magnitude is conservative (predicted 0.99–1.86 px vs actual 1.67–2.54 px) except on ARCTIC, where it is essentially calibrated (1.86 vs 1.87 px).

Diagnostics of the exact Jacobian (main paper, solver study). Under the matched budget (same energy, iterations, damping, init, and scenes, $N{=}512$ per dataset), the exact autograd fit shows 3–5× the total A-MANO joint-limit violation depth of the analytic fit (1055–1261° vs 215–551° over the 45 twist, spread, and bend axes) and a ≈2× larger fit-to-$\boldsymbol{\mu}$ residual. Two mechanisms coexist: the pose-corrective derivative lets the optimizer exploit skinning wiggle to chase 2D joints off the plausible manifold, and the exact system is worse-conditioned at the fixed iteration budget.

B.5 Inference-time anatomical-validity energy (main paper, inference-time composability)

The barrier is $E_{\text{anat}} = \frac{w}{2}\sum_{j,a}\big(\mathrm{relu}(\theta_{ja} - \theta^{+}_{ja}) + \mathrm{relu}(\theta^{-}_{ja} - \theta_{ja})\big)^2$ over the anatomy-aligned twist–spread–bend angles θ_{ja} of the 15 finger joints (A-MANO decomposition of [1]) with published per-axis limits $[\theta^-, \theta^+]$. The limits are asymmetric (a finger may flex but not hyperextend) and the twist limit is zero, so any twist is penalised. It enters the solver as one additional (detached) Gauss–Newton block in the last 5 LM iterations. The inference-time table in the main paper reports the paired base vs. $+E_{\text{anat}}$ A/B on this exact protocol. Here we give its construction, significance, and robustness.

The per-axis limits $[\theta^-, \theta^+]$ (Table 4) are those of [1]. Most are asymmetric, which is why the barrier is a two-sided hinge rather than a bound on $|\theta|$: a joint may flex to 90° but must not hyperextend past 0°.

Table 4: A-MANO per-axis joint limits used by E_{anat} (degrees), over the 15 articulated joints.

joint	twist	spread	bend
finger MCP	±0	±5	$[0, 90]$
finger PIP / DIP	±0	±0	$[0, 90]$
thumb CMC	±45	$[-15, +45]$	$[0, 45]$
thumb MCP	±0	±10	$[0, 90]$
thumb PIP	±0	±0	$[0, 90]$

Table 5: Backward-mode study (frozen front-end, 50-iteration solve run to convergence, on a single GPU).

	$\cos(\nabla_{\mathrm{unr}}, \nabla_{\mathrm{impl}})$		backward (ms)		peak mem
setting	@5it	@50it	unroll	impl	unr / impl
DexYCB (8v)	0.83	0.91	15.8	**1.1**	$O(n)$ / $O(1)$
ARCTIC (8v)	0.81	0.96	15.7	**1.3**	$O(n)$ / $O(1)$
InterHand (8v)	0.22	0.88	15.6	**1.3**	$O(n)$ / $O(1)$
InterHand (2v)	0.09	0.79	15.1	**1.2**	$O(n)$ / $O(1)$

Axes with a limit of ±0 (twist, and the PIP/DIP spread) have a degenerate box, so on those the term is a quadratic penalty on any deviation rather than a barrier. Zero twist is the anatomical target, but they do trade against the data term, which is one reason the measured gain is small.

The effect is small ($\sim$0.1 mm, up to 0.4 on OakInk) but consistent and statistically significant. A paired per-scene bootstrap (identical scenes and seed) gives ΔMPVPE$= -0.065$ mm, 95% CI $[-0.071, -0.059]$ on DexYCB (66.5% per-scene win rate), and ΔMPVPE$= -0.088$ $[-0.094, -0.082]$ on ARCTIC (n=17,359). The gain is robust to the barrier weight ($w \in [25, 100]$ a plateau, w=10 neutral, 4$\times$ no degradation), consistent with the active-set construction. The cost is 2–9$\times$ slower fitting even with the last-k schedule, so we provide it as an optional component. We report the *paired within-method* improvement, not cross-method comparisons at this level, and training with the barrier is an untested alternative, so the claim is adaptability of the released model, not superiority over retraining.

B.6 Backward-mode study: implicit vs. unrolled (main paper, solver study)

Table 5 gives the full per-setting comparison on a 50-iteration solve run to convergence. cos is the cosine similarity between the unrolled and implicit gradient w.r.t. the learned 2D evidence, at 5 and 50 iterations. Unrolled activation memory grows with the iteration count n, while implicit is $O(1)$. The implicit gradient is the convergence limit of unrolling at a fraction of the cost.

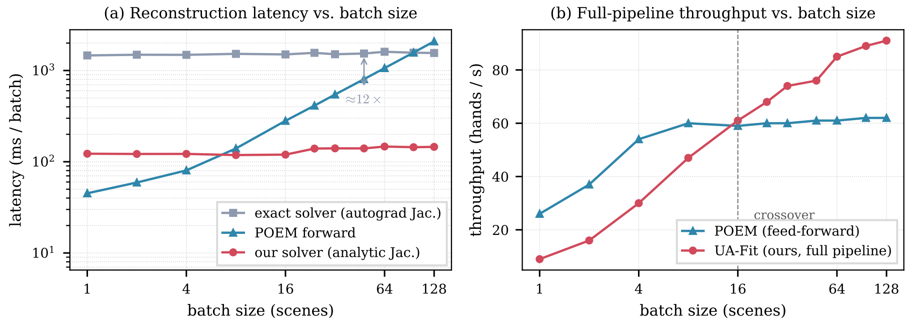

Fig. 2: Batch-size scaling on one H200 (V=8, DexYCB). **(a)** Per-batch reconstruction latency: our analytic solver and the exact autograd solver are both flat in batch size, with ours ≈12× cheaper, whereas POEM's learned fusion grows linearly. **(b)** Full-pipeline throughput: UA-Fit crosses over POEM at batch ≈16 and reaches 91 hands/s at batch 128, while POEM plateaus at ∼62.

B.7 Inference efficiency and batch scaling (main paper, efficiency)

Fig. 2 profiles the reconstruction stage on one H200 (V=8, DexYCB, CUDA-synced, warmed up, averaged). **(a)** The analytic-Jacobian solver is nearly flat in wall-clock across batch size (122 → 146 ms per batch from 1 to 128 scenes), because it batches the dense 51×51 systems into a single solve, so its per-scene cost falls almost ideally (122 → 1.1 ms). The exact autograd Jacobian is flat too but ≈12× slower at every batch size (its cost is dominated by graph construction rather than batch size, with the per-iteration ratio being the ≈14× of the main paper), while POEM's learned fusion grows linearly (per-scene flat at ∼16 ms). **(b)** End to end, UA-Fit carries a fixed LM overhead that does not amortize at batch 1 (POEM is ∼3× faster there), reaches throughput parity at batch 16 (61 vs 59 hands/s), and overtakes POEM thereafter, reaching 91 hands/s at batch 128 while the feed-forward baseline plateaus at ∼62. The efficiency result is thus throughput-at-batch: a 20× smaller reconstruction head and a solver whose cost is a batched linear solve, rather than a single-scene-latency win.

References

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